SkillNavora

Audit

Course gap audit

The master prompt checked topic by topic against the existing course, with the status of every area.

Course gap check: supplement to the Data Science, AI and Data Engineering course. Existing lessons are unchanged.

Varun Kumar · SkillNavora · skillnavora.in
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Course gap audit

This audit checks the Data Science, AI and Data Engineering master prompt against the existing course, topic by topic. Each row says whether the area is covered, partly covered or missing, and what the supplement adds.

Summary: 8 areas covered, 14 partly covered, 9 missing. The three supplements fill the missing and partial areas. Existing lessons, PDFs and tabs were not changed.

Coverage by prompt module

Prompt module	Course home	Status	What the course covers	What is missing or partial	Where the supplement covers it
1. Advanced Excel for Data Analysis	Course Module 1: Advanced Excel (12 lessons)	COVERED	Fundamentals, formatting, validation, formulas, text/logical/IF, sorting and dynamic arrays, XLOOKUP/VLOOKUP/INDEX-MATCH, pivot tables, dates and named ranges, charts, Superstore dashboard, Power Query, Power Pivot, What-If.	Missing: Excel macros and VBA, Office Scripts, Flash Fill and Text-to-Columns as named steps, cohort and customer segmentation in Excel, and the prompt's Excel projects (customer complaint analysis, financial tracker, employee productivity dashboard).	Supplement B, section B1
2. SQL for Data Analysis and Database Management	Course Module 2: SQL (15 lessons)	PARTIAL	Databases and RDBMS, keys and normalisation, DDL/DML, SELECT and filters, functions, GROUP BY/HAVING/CASE, joins, set operators, subqueries, window functions, CTEs, views, procedures, transactions, indexes, end-to-end case study.	Missing: recursive CTEs, percentile calculations, partitioning, incremental loading and SQL data-quality checks, OLTP vs OLAP in depth, and the database platforms list (SQL Server, BigQuery, Snowflake, Databricks SQL). Business SQL for CLV, churn, funnel and cohort analysis is only partly in the retail case study.	Supplement B, section B2

3. Power BI and Business Intelligence	Course Module 3: Power BI (6 lessons)	**PARTIAL**	Power BI start, Power Query and star schema, DAX fundamentals, DAX time intelligence, visuals and drill-through, service, connectivity modes, security and a capstone.	Missing: Copilot inside Power BI as a lesson, Microsoft Fabric, semantic models and deployment pipelines in depth, dataflows, incremental refresh and gateways. Industry dashboards (HR, finance, marketing, operations) are only examples.	Supplement A, section A10 (Fabric and platforms); Supplement C (dashboard interview)
4. AI and Data Analysis	Course Module 5: AI for Data Analysis (4 lessons)	**PARTIAL**	GenAI for analysts, prompting, AI with SQL and Text-to-SQL, AI in Excel and Office, the AI-assisted analyst workflow with agents and governance.	Missing: AI anomaly detection and forecasting as hands-on lessons, AI-generated DAX, and a structured tool comparison for Julius AI, NotebookLM and Perplexity.	Supplement C, section C6 (AI analytics checklist)
5. Python Programming for Data Science	Course Module 6: Python for Data Science (29 lessons)	**COVERED**	Python basics to advanced, OOP, files and JSON, environments and Git, testing and debugging, NumPy, pandas I to IV, visualisation, EDA, SQL from Python, APIs and scraping, performance, DSA, retail capstone.	Partly missing: automation of email and scheduled scripts, logging, type hints and packaging as one lesson, and SciPy and statsmodels in depth.	Supplement B, section B6 (production Python)
6. Statistics and Probability	Course Module 7: Statistics & ML (first lessons)	**COVERED**	Descriptive statistics, probability and distributions, sampling and CLT, hypothesis testing (z, t, chi-square, ANOVA, non-parametric), correlation, A/B testing and experiments.	Partly missing: business statistics for revenue forecasting and risk analysis as worked cases.	Covered by the existing lessons; Supplement C adds interview questions
7. Machine Learning and Predictive Analytics	Course Module 7: Statistics & ML (remaining lessons)	**COVERED**	Introduction to ML, scikit-learn workflow, preprocessing and features, regression, gradient descent, regularisation, logistic regression, metrics, cross-validation and tuning, KNN, trees, ensembles, SVM and Naive Bayes, imbalanced data, clustering and PCA, time series, NLP, neural networks, XAI and MLOps, churn capstone.	Missing: recommendation systems (listed as a project in the prompt) and a dedicated RMSE and R-squared review lesson.	Supplement B, section B7 (recommendation systems)

8. Data Engineering Fundamentals	NOT IN THE COURSE	**MISSING**	Only small touches: Parquet and DuckDB in the Python performance lesson, and API ingestion in the APIs lesson.	Missing: data engineering overview and roles, data formats (CSV, JSON, Parquet, Avro, ORC, compression), ingestion (batch, streaming, API, CDC), ETL vs ELT, idempotency and retries, data modelling levels, slowly changing dimensions, data quality dimensions, and the data engineering tools.	Supplement A, sections A1 to A6
9. Data Warehousing, ETL and Data Pipelines	NOT IN THE COURSE	**MISSING**	Star schema appears in the Power BI lesson only.	Missing: data warehouse, data mart, data lake and lakehouse concepts, medallion architecture, Airflow and DAGs, dbt models and tests, Spark and PySpark, Kafka and streaming, pipeline monitoring.	Supplement A, sections A3 to A8
10. Cloud Computing and Modern Data Platforms	NOT IN THE COURSE	**MISSING**	Docker is mentioned in the AI SQL lesson; Metabase is shown there too.	Missing: cloud fundamentals, AWS, Azure and Google Cloud data services, Snowflake, Databricks and Fabric platform concepts, cloud cost management, Docker, Kubernetes, CI/CD and secrets.	Supplement A, sections A9 to A11
11. Generative AI and Large Language Models	Course GenAI track (12 lessons)	**COVERED**	LLM basics, prompt engineering, calling LLM APIs, embeddings and vector search, RAG, tool use, agents, MCP, evaluation and guardrails, customising models, GenAI for data work, shipping a GenAI app.	Partly missing: context engineering as a lesson of its own, and a provider comparison table for OpenAI, Claude and Gemini with current model names and pricing (to be checked on the day of publishing).	Supplement C, section C6

12. AI Agents and Agentic Workflows	Course Agents track and GenAI lesson 7	**PARTIAL**	Agents, tool calling, MCP, no-code agents, Claude Code, voice agents, multi-agent systems in the GenAI lessons.	Missing: hands-on LangGraph and CrewAI lessons, AutoGen concepts, agent evaluation and monitoring (task completion, traces, regression tests).	Supplement C, section C5
13. MCP, APIs and Integrations for PMs and Business Stakeholders	Course Agents track (APIs, MCP, integrations)	**PARTIAL**	APIs without code, REST, JSON, authentication, webhooks, Postman, MCP without code, n8n, Make, Zapier, Replit.	Missing: OpenAPI and Swagger, OAuth 2.0 in depth, MCP security considerations, and integration examples for CRM, ERP, Jira, Slack and Notion.	Supplement B, section B8
14. AI and No-Code Tools Laboratory	Course tools track (2 lessons: Figma, n8n)	**PARTIAL**	Figma and n8n tutorials; the Claude, n8n, Make and Zapier lesson.	Missing: lessons for most tools in the prompt (Copilot, Cursor, Replit, Lovable, Bolt, v0, Windsurf, Power Automate, Google Sheets, Looker Studio, Metabase, Supabase, Firebase, Airtable, FigJam, Miro, Jira, Notion), and the tool-evaluation format (pricing, free-tier limits, alternatives).	Supplement C, section C7 (tool evaluation sheet)
15. Data Governance, Security and Responsible AI	Scattered: guardrails (GenAI), fairness and XAI (ML), RLS (Power BI), security (SQL)	**MISSING**	Guardrails, prompt injection, fairness and explainability, row-level security, transactions and security basics.	Missing as a module: data ownership and stewardship, data catalogues, lineage, classification, retention, GDPR, India's DPDP Act, data masking, privacy by design, threat modelling and incident response.	Supplement B, section B5

16. Business Problem-Solving and Analytics Consulting	NOT IN THE COURSE as one module	**MISSING**	Business framing in the ML introduction; BRD and FRD guides on the site; PM track discovery material.	Missing: business model and KPI basics as a lesson, stakeholder interviews, Five Whys and fishbone, problem statements, hypothesis development, requirements and acceptance criteria, executive communication and storytelling, industry applications list.	Supplement B, section B3 and B4
17. Capstone Projects and Portfolio Development	Course capstone lessons (one per track) and Career track (4 lessons)	**PARTIAL**	One capstone per track (Excel, SQL, Power BI, Tableau, AI, Python, ML, GenAI, agents), career portfolio and GitHub lessons.	Missing: the 8 project briefs of the prompt as one portfolio workbook with requirements, architecture templates, testing frameworks and the GitHub submission checklist. Project 3 (end-to-end data engineering pipeline) and Project 8 (industry capstone) have no course lesson.	Supplement C, section C4 (8 capstone briefs)
18. Interview Preparation and Career Readiness	Career track (4 lessons) and site interview pages	**PARTIAL**	Data analyst and data scientist interview preparation, portfolio and GitHub, 90-day plan.	Missing: data engineering interview questions, scenario-based interview answers (7 business scenarios), behavioural STAR practice sheets, the 12 career pathways with role notes, and the final career-readiness assessment.	Supplement C, sections C1 to C3

Part C: 20 cross-functional capabilities	Spread across the course	**PARTIAL**	Git and GitHub, testing and debugging, REST APIs and webhooks, DSA, Python production topics, MLOps (in the ML lessons), LLM evaluation.	Missing: Linux and command line (only basic terminal use in Python lessons), system design fundamentals, orchestration (Airflow), dbt, data observability, cloud cost optimisation, data and AI architecture, technical documentation, Agile and project delivery, production troubleshooting, business communication.	Supplement A sections A4 and A12; Supplement B sections B5, B9 and B10
Part D: 12-month roadmap	Career lesson 4 (90-day plan) only	**PARTIAL**	A 90-day plan in the career lessons.	Missing: the 12-month roadmap with weekly hours, milestones and assessment criteria for each stage.	Supplement C, section C8
Part E: Tools & Learning Resources directory	Site: Tools & learning resources page	**PARTIAL**	The site has a tools and learning resources page with categorised links.	Not yet organised by the 16 categories of the prompt, and the links are not all date-checked (the prompt requires a checked date).	Supplement C, section C7 (categories and checked dates)
Part G: PDF 1 Master Handbook	Course PDFs: Complete Data Science & AI Handbook Parts 1 to 4	**PARTIAL**	Four complete handbook parts exist.	Needs a check that the parts include every module above (Data Engineering and Governance are not in them).	Supplement A, Supplement B and this audit complete the set
Part G: PDFs 2 to 6 (Excel, SQL, Power BI, Python, Stats & ML)	Course lesson PDFs by track	**COVERED**	One PDF per lesson in each track (Excel 12, SQL 15, Power BI 6, Python 29, Stats & ML 28).	Track-level handbooks combining the lessons are not separate PDFs.	Existing lesson PDFs are unchanged
Part G: PDF 7 Data Engineering and Cloud Handbook	NOT IN THE COURSE	**MISSING**	None.	The whole handbook.	Supplement A (full handbook)
Part G: PDF 8 to 9 (GenAI, RAG, Agents and MCP)	Course GenAI and Agents lesson PDFs	**COVERED**	Lesson PDFs for GenAI (12) and Agents (6).	Track-level handbooks are not separate PDFs.	Existing lesson PDFs are unchanged

Part G: PDF 10 AI and No-Code Tools Laboratory	Tools track (2 lessons)	**PARTIAL**	Figma and n8n.	The tool directory and setup guides for the other tools.	Supplement C, section C7
Part G: PDF 11 Capstone Projects and Portfolio Workbook	Career lessons and capstone lessons	**PARTIAL**	Capstone lessons and portfolio lessons.	The workbook with all 8 briefs, templates and rubrics.	Supplement C, section C4
Part G: PDF 12 Interview Preparation Handbook	Interview-Handbook.pdf and career interview lessons	**COVERED**	Interview handbook and two career interview lessons.	Data engineering and scenario sections.	Supplement C, sections C1 to C3
Part G: PDF 13 12-Month Roadmap and Tracker	NOT IN THE COURSE	**MISSING**	None as a PDF.	The whole roadmap and tracker.	Supplement C, section C8
Part G: PDF 14 Quick Reference and Cheat Sheets	NOT IN THE COURSE	**MISSING**	None as one collection.	Excel, SQL, DAX, Python, pandas, statistics, ML metrics, API and MCP terms, AI terms.	Supplement B, section B11 (cheat sheet index)
Part G: PDF 15 Data Science and AI Glossary	NOT IN THE COURSE	**MISSING**	None as one glossary.	Terms and acronyms with simple definitions and examples.	Supplement B, section B11 (glossary index)

Proposed additions for your approval

These are added as supplements, not as edits to existing lessons. You approve each one before any change to the course itself.

- Add a Data Engineering module (Supplement A) as course Module 8 with hands-on labs (Airflow, dbt, Docker, one cloud).
- Add a Governance and Responsible AI module (Supplement B, B5) and a Business Consulting module (B3, B4).
- Add lessons for Excel automation (B1) and the missing SQL topics (B2).
- Add the eight capstone briefs (C4) as a workbook, and the 12-month roadmap (C8) as a tracker.
- Add the no-code tool lessons (Part E of the prompt) in the tools track.